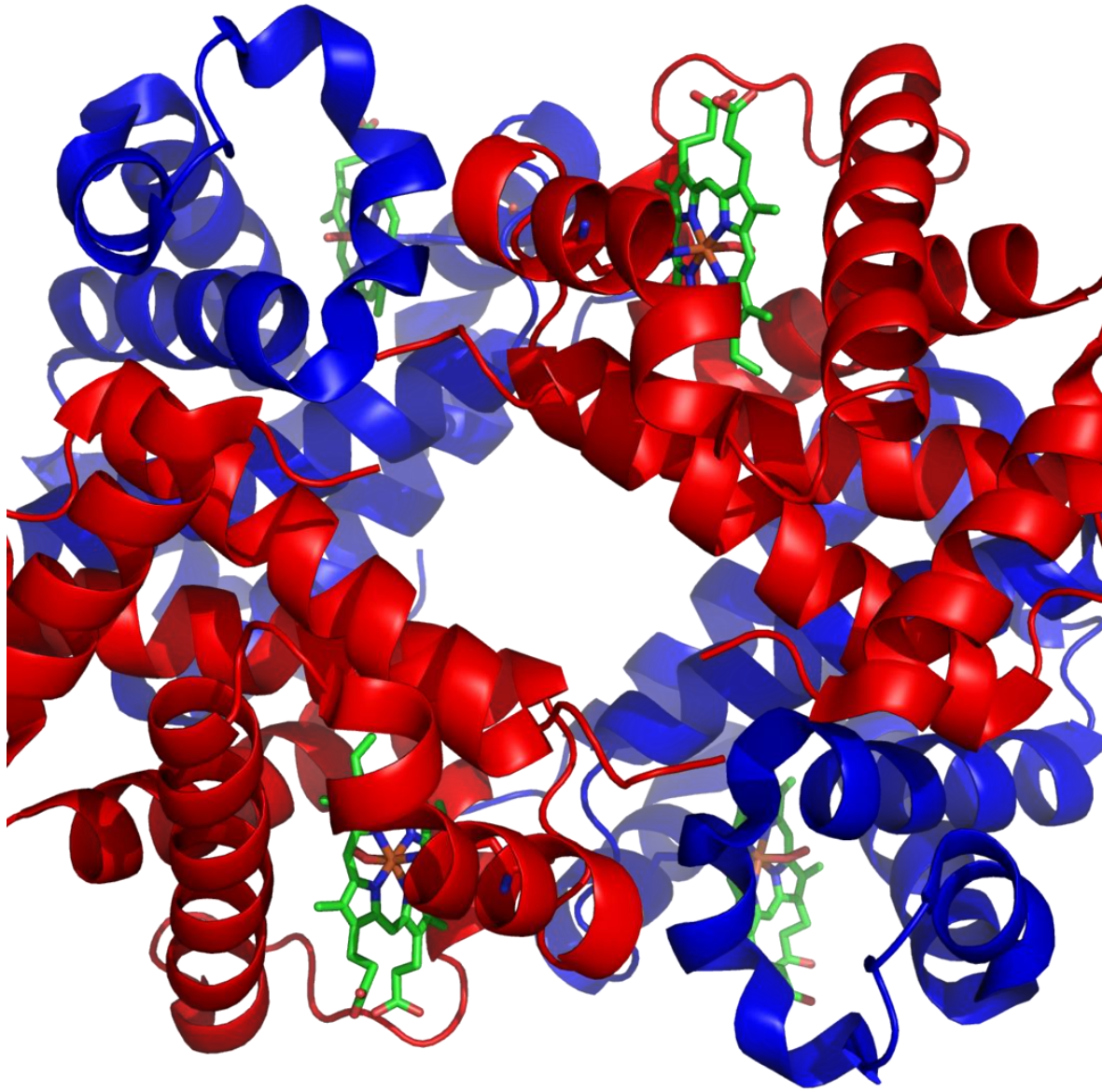
DP Biology – B3.1 (HL extension)

Haemoglobin and oxygen

- What adaptations let multicellular organisms exchange gases?
- How do flowering plants and mammals differ, and resemble each other, in gas exchange?



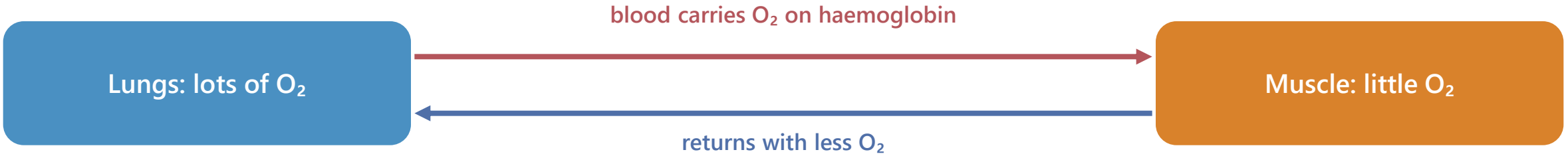
HL statements B3.1.11–13. Textbook pp. 292–294.
Image: haemoglobin, four subunits with their haem groups. Zephyris, CC BY-SA 3.0.

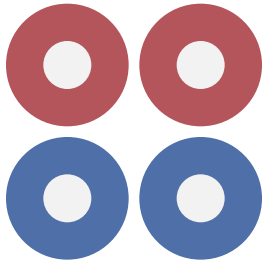
Objectives for each class

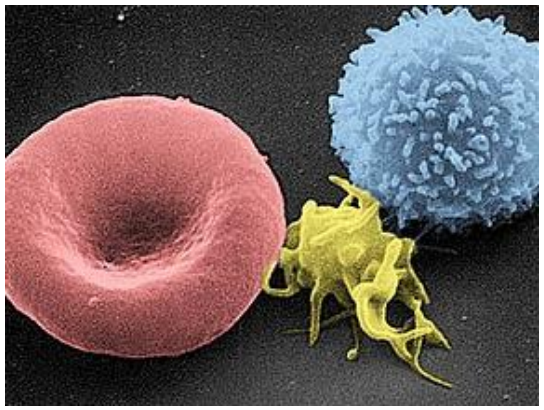
Class	Statements	By the end of the class, you can...
1	B3.1.13	Explain partial pressure, read oxygen dissociation curves, and explain their S shape
2	B3.1.11	Compare foetal and adult haemoglobin, including allosteric binding of CO_2
3	B3.1.12	Explain the Bohr shift and how it helps actively respiring tissues

Load in the lungs, unload in the tissues

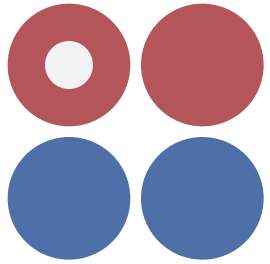
B3.1.11–13





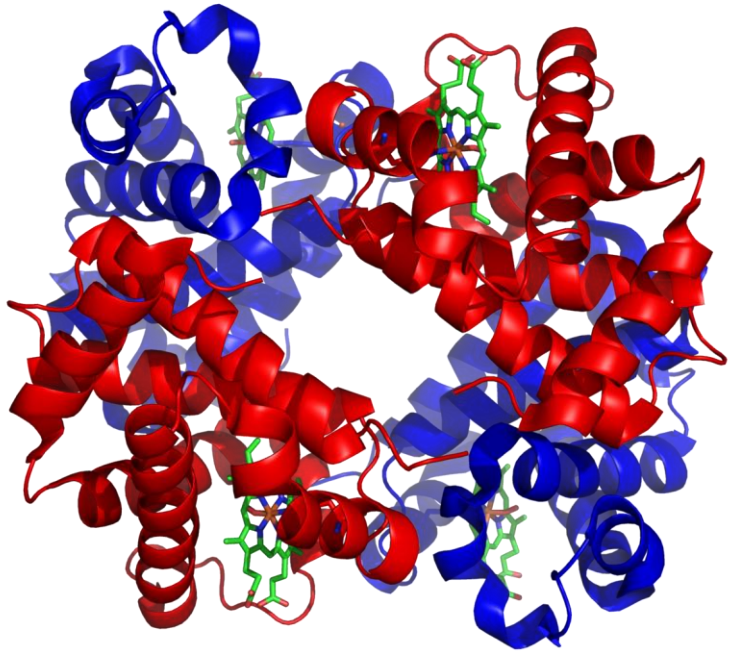


Red blood cell. NCI, public domain.

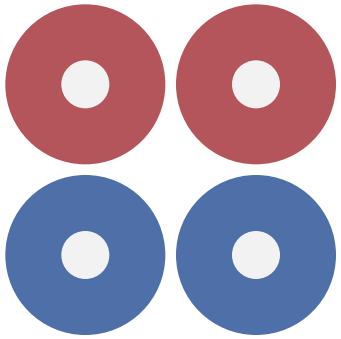


A good oxygen carrier must do two opposite things: **grab** oxygen where there is plenty, and **let go** of it where there is little. Haemoglobin does both, and it adjusts to how hard the tissue is working.

Haemoglobin



Haemoglobin. Zephyris, CC BY-SA 3.0.



alpha chains (red), beta chains (blue)

Four polypeptide chains, each with a **haem group** containing **iron**.

Each haem binds one O_2, so one haemoglobin carries up to **four**.

Binding is **reversible**: oxygen loads and unloads.

Link

A protein with quaternary structure and a non-protein part (haem): a conjugated protein (B1.2).

Partial pressure of oxygen

B3.1.13

Place	pO_2 / kPa
Air at sea level	21
Alveoli	13
Tissues at rest	5
Muscle in hard exercise	1–2

Air is a mix of gases. The **partial pressure of oxygen (pO_2)** is the part of the total pressure due to oxygen, in kPa.

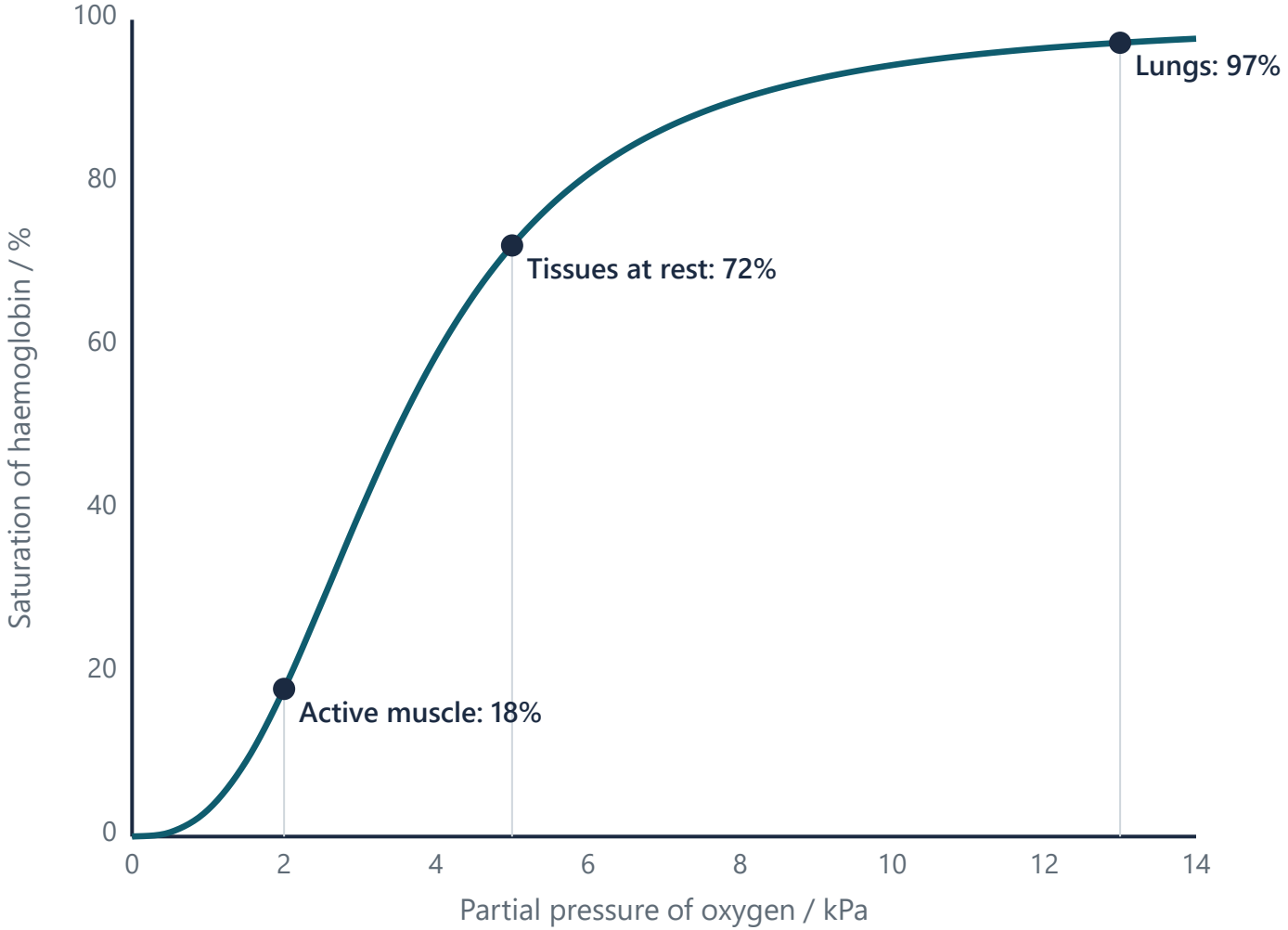
Oxygen always diffuses from high pO_2 to low pO_2: alveoli → blood → tissues.

Think

Why is pO_2 in the alveoli lower than in the air you breathe in?

The oxygen dissociation curve

B3.1.13



Adult haemoglobin. Modelled curve with typical values.

The curve shows how full haemoglobin is (% saturation) at each pO_2.

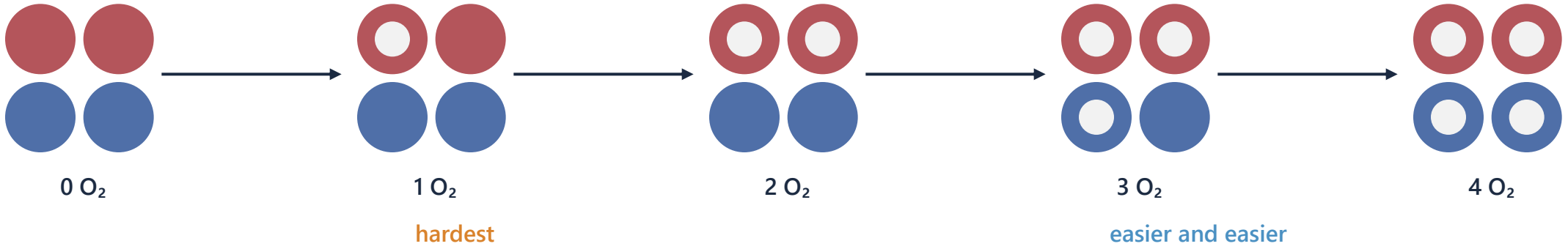
It is **S-shaped**. It is steepest at 2–6 kPa, the range in tissues, so a small fall in pO_2 releases a lot of oxygen.

Try it

How much oxygen is released between lungs and resting tissue?

Why the curve is S-shaped

B3.1.13

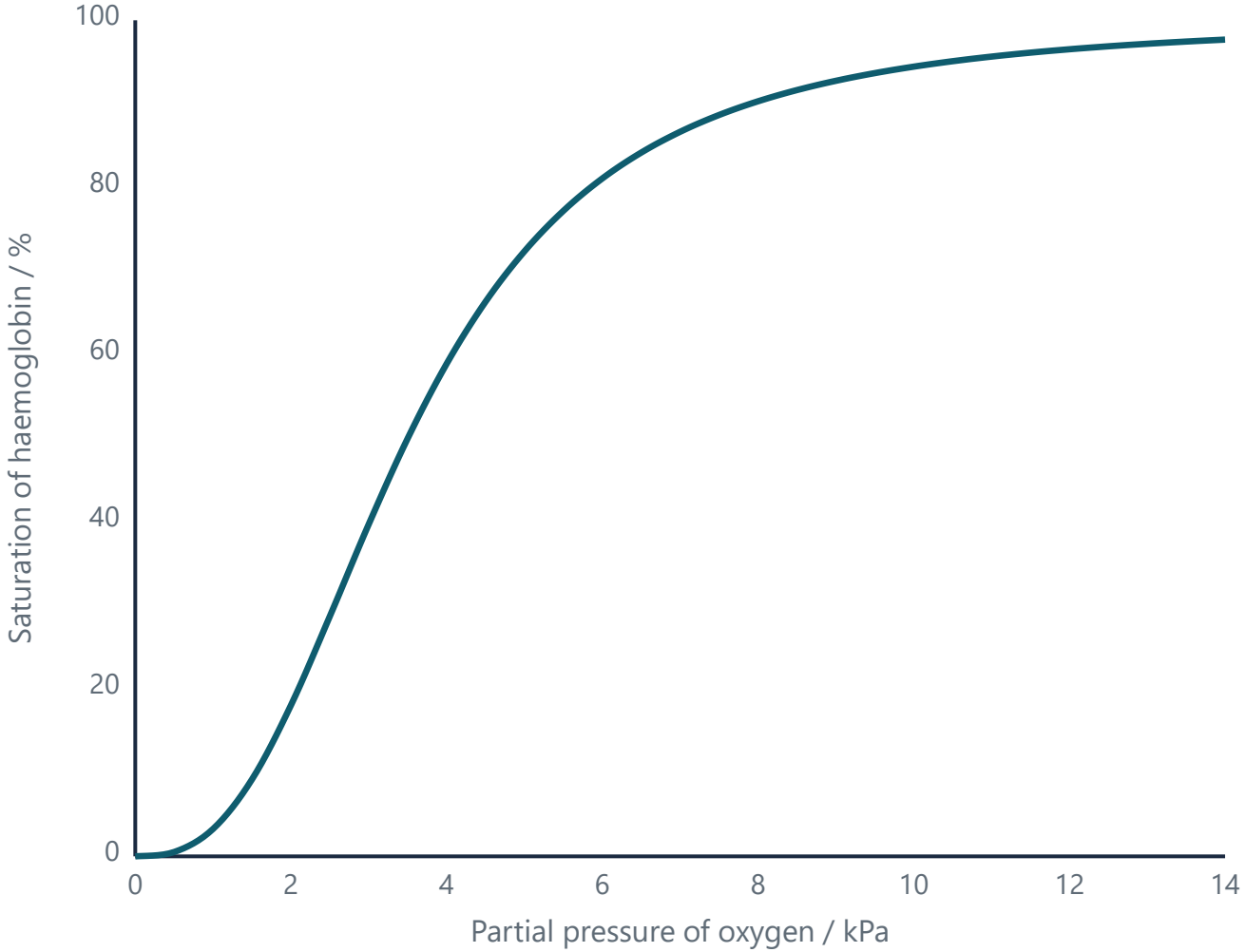


The first O_2 binds with difficulty. It changes the shape of its subunit, and that change passes to the others, raising their **affinity**: **cooperative binding**.

So the curve starts flat, then climbs steeply, then levels off when almost full.

It works in reverse too: once one O_2 leaves, the others leave more easily. Great for unloading in tissues.

Practice: reading the curve



Adult haemoglobin. Modelled curve.

1. Read the saturation at 3 kPa and at 10 kPa.
2. At what pO_2 is haemoglobin 50% saturated?
3. Blood leaves the lungs at 97% saturation and enters a muscle at 3 kPa. What percentage of the oxygen is released?

Working with data: oxygen delivery

B3.1.13

Oxygen saturation of blood entering and leaving a leg muscle. These values are invented for practice; the pattern is realistic.

Activity	Entering / %	Leaving / %
Rest	97	75
Walking	97	55
Running	97	25

1. Calculate the percentage of oxygen unloaded in each case.
2. Explain why unloading increases with exercise.

Check-in: class 1

B3.1.13

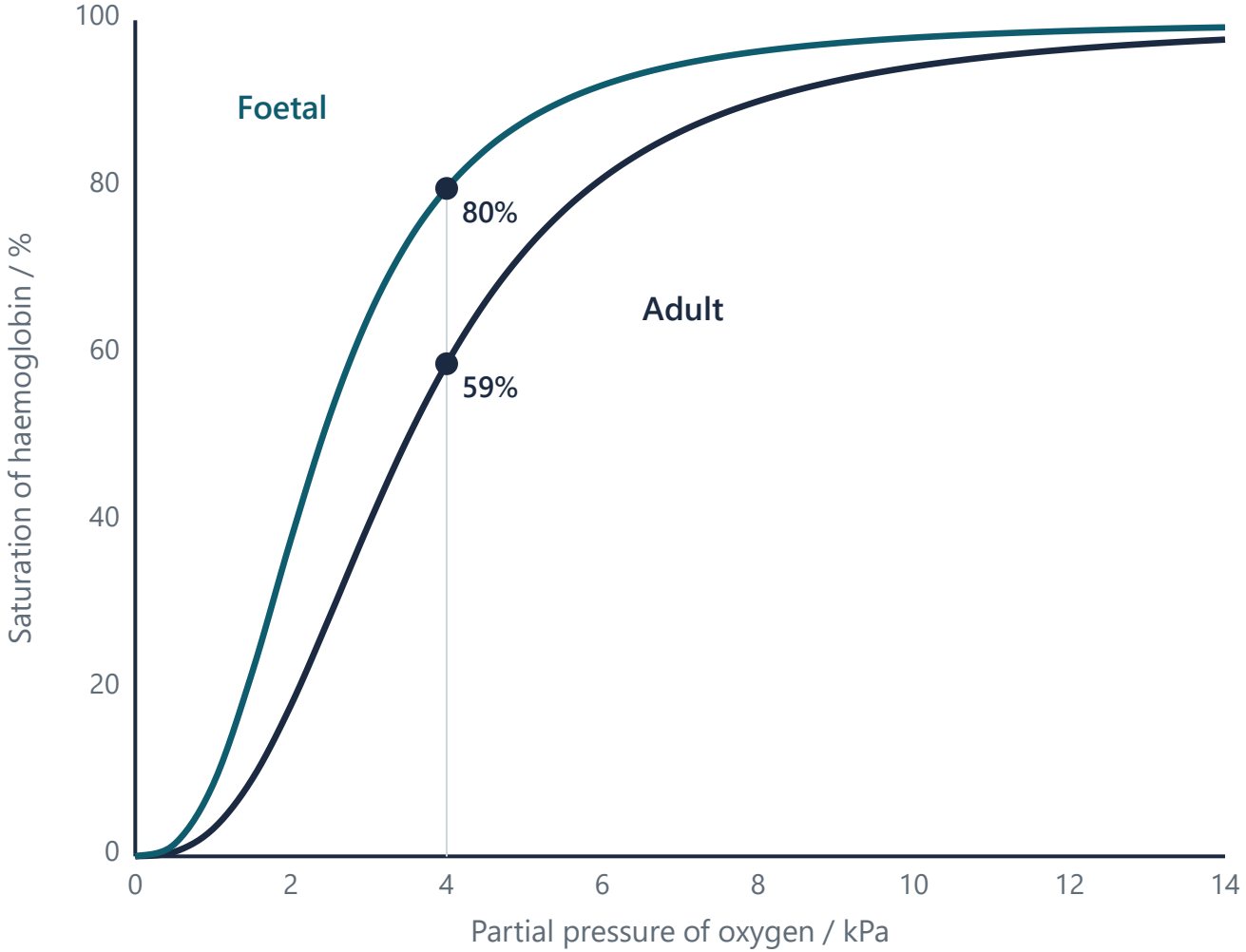
1. State how many O_2 molecules one haemoglobin can carry.
2. Define partial pressure of oxygen.
3. Describe the shape of the oxygen dissociation curve.
4. Explain the S shape using cooperative binding.
5. Explain why the steep part of the curve matters in tissues.

How to use this

Answer without your notes, then check against slides 4–9.

Foetal and adult haemoglobin

B3.1.11



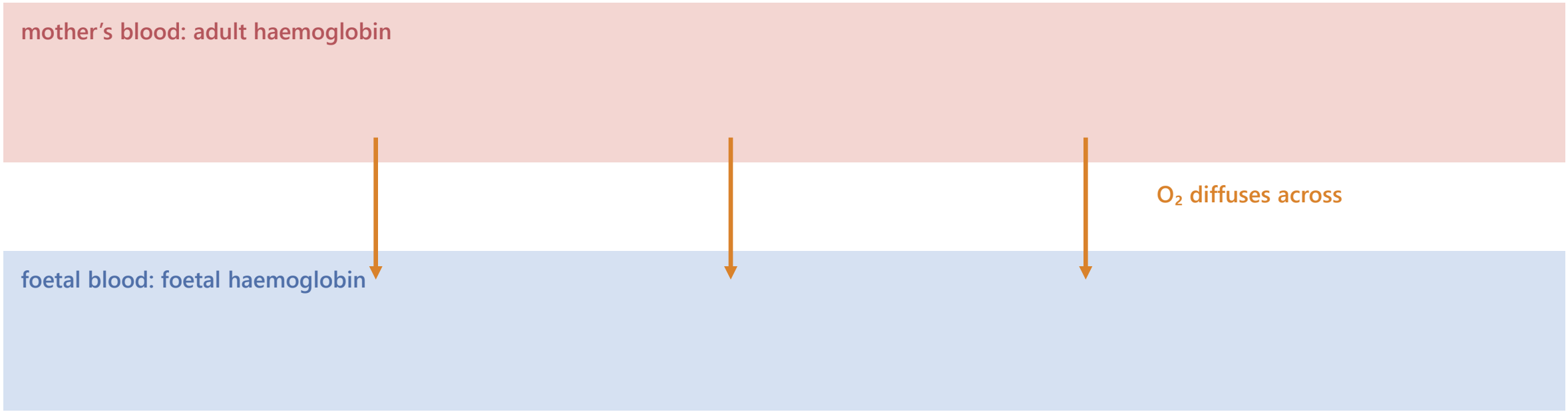
Modelled curves with typical values.

Foetal haemoglobin has a **higher affinity** for oxygen: its curve is to the **left**.

At 4 kPa in the placenta: foetal 80%, adult 59%. So oxygen passes from mother to foetus.

Why the foetus needs it

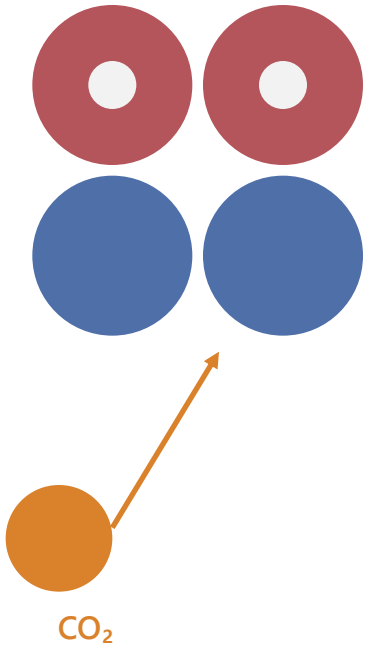
B3.1.11



The foetus has no working lungs: all its oxygen comes from the mother's blood in the placenta. The two bloods come close but never mix.

Because foetal haemoglobin binds oxygen more strongly, it pulls oxygen away from the mother's haemoglobin.

An allosteric protein



Haemoglobin is **allosteric**: molecules binding at sites other than the haem can change its shape.

CO_2 binds allosterically and **lowers** the affinity for oxygen, so more oxygen is released.

This effect is weaker for foetal haemoglobin, another reason it holds oxygen more strongly.

Link

Allosteric sites in enzymes: C1.1 HL (allosteric inhibition).

Living high up



A llama. Johann Dreo, CC BY-SA 3.0.

At high altitude pO_2 is low, so loading oxygen in the lungs is harder.

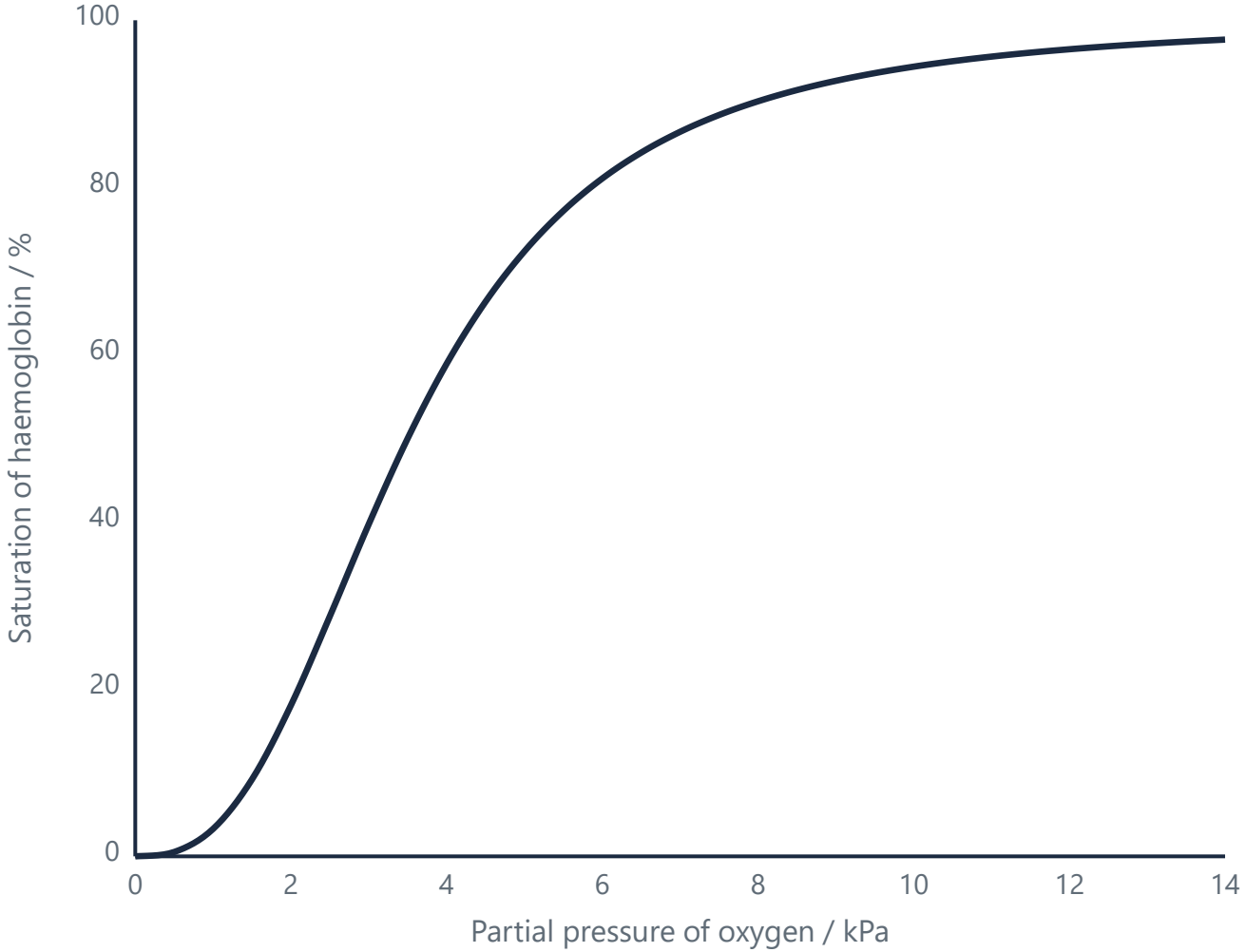
Llamas and vicunas have haemoglobin with a **higher affinity**: their curve is shifted to the left, like foetal haemoglobin.

Think

Mountain climbers produce more red blood cells over a few weeks. Why would this help when the curve itself cannot change?

Practice: sketch the curves

B3.1.11



Adult haemoglobin. Modelled curve.

1. Copy the adult curve.
2. Add the foetal haemoglobin curve and label it.
3. Add a curve for llama haemoglobin.
4. Explain the position of each curve.

Check-in: class 2

B3.1.11

1. Compare the affinity of foetal and adult haemoglobin.
2. Explain why foetal haemoglobin needs a higher affinity.
3. Define an allosteric protein.
4. State the effect of CO_2 binding on haemoglobin.
5. Suggest why animals at high altitude have high-affinity haemoglobin.

How to use this

Answer without your notes, then check against slides 11–15.

The Bohr shift

Respiring cells release CO_2 → CO_2 enters red blood cells and forms carbonic acid → pH falls (more H^+) → H^+ and CO_2 change haemoglobin's shape → **Affinity falls: more O_2 released**

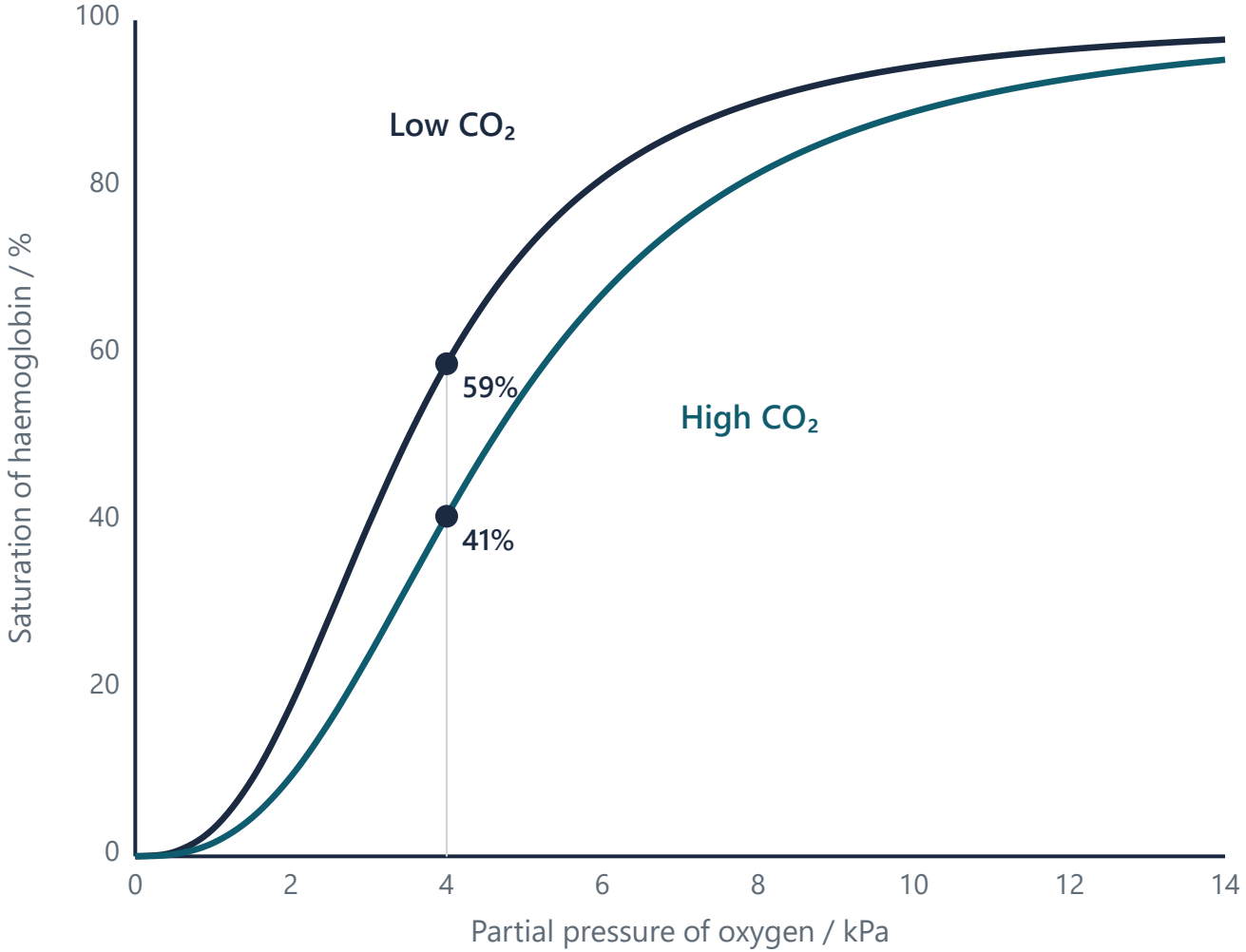
More CO_2 shifts the curve to the **right**. At the same pO_2, haemoglobin is less saturated, so it gives up **more oxygen** exactly where respiration is fastest.

In the lungs, CO_2 leaves, pH rises, and affinity goes back up for loading.



Athlete training. Florida Memory, public domain.

Reading the Bohr shift



Modelled curves with typical values.

At 4 kPa: low CO_2 59%, high CO_2 41%. So an extra **18%** of the oxygen is released in tissue rich in CO_2.

At high pO_2 the curves almost meet, so loading in the lungs is hardly affected.

Left and right shifts

B3.1.11–12

Shift to the left: higher affinity	Shift to the right: lower affinity
Foetal haemoglobin	More CO_2 (Bohr shift)
High-altitude animals (llama)	Lower pH
Low CO_2, as in the lungs	Higher temperature, as in working muscle

Rule of thumb

Left = loads more easily (good where oxygen is scarce to pick up). Right = unloads more easily (good where oxygen is needed).

Working with data: CO_2 and saturation

B3.1.12

Saturation of haemoglobin at 4 kPa oxygen with different partial pressures of CO_2. These values are invented for practice; the pattern is realistic.

pCO_2 / kPa	Saturation / %
3	64
5	59
7	50
9	41

1. Describe the effect of CO_2 on saturation.
2. Explain the effect.
3. Suggest why this helps a muscle during exercise.

Practice: apply the ideas

B3.1.11–13

A baby is born

Predict what happens to its foetal haemoglobin over the next months, and why.

A cyclist sprints uphill

Explain two reasons the leg muscles receive more oxygen.

A climber arrives at 5000 m

Explain why loading oxygen in the lungs is harder.

How to use this

Answer in 2–3 sentences each, using the words affinity, partial pressure and curve.

Common mistakes to avoid

Not this	But this
A right shift means more oxygen is carried.	A right shift means more oxygen is released.
Foetal haemoglobin has a lower affinity.	It has a higher affinity: curve to the left.
CO_2 binds to the haem group.	CO_2 binds allosterically, away from the haem.
The curve is a straight line.	It is S-shaped because of cooperative binding.

Check that you can...

1. **Define** partial pressure of oxygen
2. **Read** saturation values from oxygen dissociation curves
3. **Explain** the S shape using cooperative binding
4. **Compare** foetal and adult haemoglobin
5. **Explain** the allosteric effect of CO_2
6. **Explain** the Bohr shift and its benefit to respiring tissue

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. State what causes the oxygen dissociation curve to be S-shaped.
2. Explain why foetal haemoglobin needs a higher affinity for oxygen.
3. Describe the effect of a rise in CO_2 on the curve and on oxygen delivery.

Image credits

1GZX Haemoglobin.png – Zephyris at English Wikipedia – CC BY-SA 3.0

Red White Blood cells.jpg – Electron Microscopy Facility at The National Cancer Institut – Public domain

Llama lying down.jpg – Johann "nojhan" Dréo – CC BY-SA 3.0

FAMU athlete Robert Hayes practices running on the track.jpg – Florida Memory – Public domain

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